

Generation Interconnection Request System Impact Study Report

For Queue Project Q20-1503

Pheasant Branch 138 kV
April 19, 2022

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q20-1503. The Interconnection Customer proposes a 559.6 MW natural gas combined cycle facility interconnecting at the Pheasant Branch 138 kV substation in the Virginia area. The request was received on June 23, 2020. The study evaluated system intact and single contingency conditions using the regional planning case, including 407 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 25 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$28,014,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
SWITZER – PLAZA 138 KV Line	IMPERIAL SAVANNAH LP – PHEASANT BRANCH 138 KV	160.8	750	100.0
IMPERIAL SAVANNAH LP – HAGGIN 138 KV	PHEASANT BRANCH – SWITZER 138 KV	158.7	775	100.0
IMPERIL SAVANNAH LP – PHEASANT BRANCH 138 KV	PHEASANT BRANCH – SWITZER SUBSTATION 138 KV	157.3	792	100.0
PHEASANT BRANCH – SWITZER 138 KV Line	IMPERIAL SAVANNAH LP – PHEASANT BRANCH 138 KV Line	157.3	792	100.0
FRITCH – RIV ROAD 138 KV	FRITCH – LAKESIDE 138 KV Line	153.7	208	22.5
PLAZA SUBSTATION – KEVLAR 138 KV	PLAZA – BIG HANAFORD 138 KV Line	143.6	248	29.6
RIV RD SUBSTATION – HUMACAO 2601 SUBSTATION 138 KV	LAKESIDE – FRITCH 138 KV	132.6	224	22.5
PLAZA – BIG HANAFORD 138 KV	IMPERIAL SAVANNAH LP – PHEASANT BRANCH 138 KV	131.9	539	45.9
PARAGOULD RECIPROCATING – KEVLAR 138 KV L	PLAZA – BIG HANAFORD 138 KV	130.9	237	17.8
PARAGOULD RECIPROCATING – MOONSTONE 138 KV Line	BIG HANAFORD – PLAZA 138 KV	123.4	217	17.8
CENTRALIA – KING CITY 138 KV	BIG HANAFORD – FRITCH 138 KV	122.5	353	22.5
CARVER – BIG HANAFORD 138 KV line	PLAZA – BIG HANAFORD 138 KV	120.0	359	22.2
CHANCELLOR ROW – NATCHEZ 138 KV	SKAGGS ISLAND JUNCTION 2 – MITCHELL DAM 345 KV line	117.9	367	18.9

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
CHANCELLOR ROW – SHACK MILLS STA 138 KV	CHANCELLOR ROW – BEAR ISLAND 345 KV	117.4	284	7.9
NATCHEZ – LAKESIDE 138 KV line	MITCHELL DAM – SKAGGS ISLAND JCT 2 345 KV	116.4	390	18.9
FRITCH – BIG HANAFORD 138 KV line	MITCHELL DAM – SKAGGS ISLAND JCT 2 345 KV line	116.1	640	34.2
CARVER – SHOCKOE 138 KV	PLAZA – BIG HANAFORD 138 KV	113.7	444	22.2
HAGIN – TYLER 138 KV	PHEASANT BRANCH SUBSTATION – SWITZER 138 KV	113.4	452	29.2
FRITCH – LAKESIDE 138 KV line	MITCHELL DAM – SKAGGS ISLAND JCT 2 345 KV line	112.7	466	18.9
SHOCKOE – MANCHESTER 138 KV	BIG HANAFORD SUBSTATION – PLAZA 138 KV	112.3	370	17.9
DPS OCEAN SPRAY – FOUR RIVERS 138 KV	LEAVITT SUBSTATION – BIRCHWOOD 345 KV	111.2	329	5.2
HAGGIN 345/138kV Autotransformer	N TOWER TAP – HAGGIN 345 KV	110.6	335	17.7
TURNER ROAD – ELKO 138 KV Line	MITCHELL DAM – HAGGIN 345 KV L	106.7	403	11.0
FOUR RIVERS – TRENTON DIESEL 138 KV	LEAVITT – BIRCHWOOD SUBSTATION 345 KV	106.7	343	5.2
SHACK MILLS STATION – KIMBERLEY 138 KV	CHANCELLOR ROW SUBSTATION – BEAR ISLAND 345 KV Line	105.2	287	7.9

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
SWITZER – PLAZA 138 KV L	Rebuild line with higher-capacity conductor	\$1,366,000	44% (\$597,000)
HAGGIN SUBSTATION – IMPERIAL SAVANNAH LP 138 KV line	Rebuild line with higher-capacity conductor	\$13,014,000	44% (\$5,684,000)
IMPERIL SAVANNAH LP – PHEASANT BRANCH 138 KV	Rebuild line with higher-capacity conductor	\$3,256,000	44% (\$1,422,000)
SWITZER – PHEASANT BRANCH 138 KV line	Rebuild line with higher-capacity conductor	\$2,532,000	44% (\$1,106,000)
RIVER RD – FRITCH 138 KV	Rebuild line with higher-capacity conductor	\$4,060,000	30% (\$1,232,000)
PLAZA – KEVLAR 138 KV Line	Rebuild line with higher-capacity conductor	\$2,192,000	41% (\$893,000)
RIV RD SUBSTATION – HUMACAO 2601 SUBSTATION 138 KV	Rebuild line with higher-capacity conductor	\$1,652,000	30% (\$501,000)
PLAZA – BIG HANAFORD 138 KV	Rebuild line with higher-capacity conductor	\$7,830,000	38% (\$3,016,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
KVLAR – PARAGOULD RECIPROCATING 138 KV	Rebuild line with higher-capacity conductor	\$1,676,000	36% (\$601,000)
PARAGOULD RECIPROCATING – MOONSTONE 138 KV	Rebuild line with higher-capacity conductor	\$2,200,000	36% (\$789,000)
CENTRALIA – KING CITY 138 KV	Rebuild line with higher-capacity conductor	\$7,153,000	23% (\$1,672,000)
CARVER – BIG HANAFORD 138 KV line	Rebuild line with higher-capacity conductor	\$2,467,000	26% (\$636,000)
CHANCELLOR ROW – NATCHEZ 138 KV	Rebuild line with higher-capacity conductor	\$2,496,000	16% (\$412,000)
CHANCELLOR ROW – SHACK MILLS STA 138 KV	Rebuild line with higher-capacity conductor	\$2,494,000	17% (\$421,000)
LAESIDE SUBSTATION – NATCHEZ 138 KV	Rebuild line with higher-capacity conductor	\$2,556,000	16% (\$422,000)
FRITCH – BG HANAFORD 138 KV	Rebuild line with higher-capacity conductor	\$3,672,000	19% (\$694,000)
CARVER – SHOCKOE 138 KV	Rebuild line with higher-capacity conductor	\$1,987,000	26% (\$512,000)
HAGGIN – TYLER 138 KV	Rebuild line with higher-capacity conductor	\$477,000	30% (\$144,000)
FRITCH – LAKESIDE 138 KV L	Rebuild line with higher-capacity conductor	\$2,685,000	16% (\$443,000)
MNCHESTER – SHOCKOE 138 KV Line	Rebuild line with higher-capacity conductor	\$2,082,000	38% (\$794,000)
DPS OCEAN SPRAY – FOUR RIVERS 138 KV	Rebuild line with higher-capacity conductor	\$3,030,000	17% (\$504,000)
HAGGIN 345/138kV Autotransformer	Replace transformer with higher-rated unit	\$19,915,000	22% (\$4,457,000)
TURNER ROAD – ELKO 138 KV	Reconductor line	\$1,782,000	18% (\$325,000)
FOUR RIVERS – TRENTON DIESEL 138 KV	Reconductor line	\$3,426,000	17% (\$570,000)
SHACK MILLS STATION – KIMBERLEY 138 KV	Reconductor line	\$990,000	17% (\$167,000)
	Total allocated to project		\$28,014,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.